Containers

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Accessibility Notice

This document is also available in HTML format at:

Link: `https://aholdengouveia.name/LinuxAdmin/labexcercises/containers.html`

The HTML version provides enhanced accessibility features including keyboard navigation, screen reader support, responsive design, dark mode support, and high contrast options.

Objectives:

- The objective of this lab is to introduce students to containers, including Docker and Kubernetes, which are widely used in modern server administration and application deployment. Students will install and configure container tools, deploy a containerized web server, and evaluate the pros and cons of containers for a business.

Background

Acme Corp is considering containers for future deployments, and your manager wants a proof of concept and a recommendation. You may do this on one of the two servers you set up in the Server Setup lab.

Requirements

1. Install Docker and Kubernetes on your server. You can use MicroK8s as your Kubernetes installation method.
2. Go through the minikube tutorial.
3. Get a basic web server working using Docker or Kubernetes, your choice. You can use the Apache (httpd) Docker image, or if you would prefer, you can substitute another server such as Home Assistant or ZoneMinder. Take a screenshot showing your web server running and its page loading.
4. Write an executive summary on what containers are, the pros and cons of containers, and why you think Acme Corp should or shouldn't consider using them going forward.
5. Find at least one resource for containers that you found helpful and that I haven't already shared anywhere.

Acceptance Criteria

- The install process is documented with screenshots, explanations of what you're doing, and any troubleshooting that was needed.
- The web server is proven to work with a screenshot.
- The executive summary is less than 1 page.
- Your resource is new, not something I've already shared.
- Every terminal screenshot shows your prompt with your username and hostname, and either the output of `date` or a note beside the terminal with your name and the term. Hostnames must match your server inventory from the Server Setup lab.

Deliverables

Audience: someone technically inclined but unfamiliar with virtualization. Include screenshots to walk them through the install process.

1. A document on how to install Docker and Kubernetes on your server.
2. A short document describing how the minikube tutorial went, what gave you trouble (if anything), and how you fixed it. Include screenshots showing the steps you did and minikube up and running.

3. A screenshot of your working web server, with a short explanation of how you set it up.
4. Your executive summary.
5. A link to your container resource.

Resources

References, a video, a PowerPoint and some notes are available at my website Link: `https://www.aholdengouveia.name/LinuxAdmin/containers.html`

MicroK8s installation instructions Link: `https://microk8s.io/`

The minikube tutorial Link: `https://kubernetes.io/docs/tutorials/hello-minikube/`

The Apache (httpd) Docker image Link: `https://hub.docker.com/_/httpd`

My instructions and sample yaml files Link: `https://www.aholdengouveia.name/SmartHome/Virtualizationsetup.html`